Value putting:

i) No. of variables > no. of eqns.

ii) Formula asked $\begin{cases} v \to v \\ v \to c \end{cases}$

$\theta \to 90°/0°,\ 45°,\ 30°/60°$

$\frac{\theta}{2}, 2\theta \to 60, 30, 15$

iii) Misc.
- Putting all angles equal. (Symmetrical)
- Substituting complementary values.

1. The simplified value of $(\text{secA-cosA})^2 + (\text{cosecA-sinA})^2 - (\text{cotA-tanA})^2$ $(secA - cosA)^2 + (cosecA - sinA)^2 - (cotA - tanA)^2$ को सरल करें:

a) 0 b) $\frac{1}{2}$

c) 1 d) 2

② The value of $(1 + cot\theta - cosec\theta)(1 + tan\theta + sec\theta)$ is equal to

$(1 + cot\theta - cosec\theta)(1 + tan\theta + sec\theta)$ का मान है:

a) 1 b) 2

c) 0 d) -1

(3) If $(sin\alpha + cosec\alpha)^2 + (cos\alpha + sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, then the value of k is

अगर $(sin\alpha + cosec\alpha)^2 + (cos\alpha + sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, है, तो k का मान:

a) 1 b) 7

c) 3 d) 5

(4) Find the value of $3(\sin^4\theta + \cos^4\theta) + 2(\sin^6\theta + \cos^6\theta) + 12\sin^2\theta\cos^2\theta$

$3(\sin^4\theta + \cos^4\theta) + 2(\sin^6\theta + \cos^6\theta) + 12\sin^2\theta\cos^2\theta$ का मान क्या है|

a) 3 b) 2

c) 0 d) 5

(5) The simplified value of $(\sec x \sec y + \tan x \tan y)^2 - (\sec x \tan y + \tan x \sec y)^2$

$(\sec x \sec y + \tan x \tan y)^2 - (\sec x \tan y + \tan x \sec y)^2$ को सरल करें|

a) -1 b) 0

c) $\sec^2 x$ d) 1

6 What is the value of $\frac{[(\sin x+\sin y)(\sin x\cdot\sin y)]}{(\cos x+\cos y)(\cos y\cdot\cos x)}$?

$\frac{[\sin x+\sin y \quad \sin x\cdot\sin y]}{(\cos x+\cos y)(\cos y\cdot\cos x}$ का मान क्या है?

a) 0 b) 1

c) -1 d) 2

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7 What is the value of sin(B-C) cos(A-D) + sin(A-B)cos(C-D) + sin(C-A)cos(B-D)?

$sin(B-C)\cos(A-D)+\sin(A-B)\cos(C-D)+\sin(C-A)\cos(B-D)$ का मान क्या है?

a) $\frac{3}{2}$ b) -3

c) 1 d) 0

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⑧ What is the value of cos(90· Bs)inC·(A+) sin(90+ Acos)B+(C · sin)90·Cc(osA+ B? ( )

$cos(90-B)\,sin(C-A)+sin(90+A)\,cos(B+$

C) sin(90· C) cos(A+ B)) का मान क्या है?

a) 1 b) sinA+ B· C)

c) 0 d) cos(B+ C· A)

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(9) What is the value of

$$\frac{[\sin y\cdot z+\sin y+z+2\sin y]}{\sin x\cdot z+\sin x+z+2\sin x}$$?

$$\frac{[\sin y\cdot z+\sin y+z+2\sin y]}{\sin x\cdot z+\sin x+z+2\sin x}$$ का मान क्या है?

a) cosxsiny

b) $\frac{\sin y}{\sin x}$

c) sinz

d) sinxtany

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10. What is the value of $[(\text{Cos}^3 2\theta + 3\,Cos\,2\theta) \div (\text{Cos}^6\,\theta - Sin^6\theta)]$?

$[(\text{Cos}^3 2\theta + 3\,Cos\,2\theta) \div (Cos^6\,\theta - Sin^6\theta)]$ का मान क्या है?

a) 0 b) 1

c) 4 d) 2

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(11) $\frac{\cot\theta}{\cot\theta - \cot 3\theta} + \frac{\tan\theta}{\tan\theta - \tan 3\theta} = ?$

a) 1 b) -1

c) 2 d) 0

(12) The value of $\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} \times \frac{\tan^2\theta(\text{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$ is:

$\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} \times \frac{\tan^2\theta(\text{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$ का मान क्या है?

a) 0 b) -1

c) 1 d) $\frac{1}{2}$

SSC CGL 2018 T-2

(13) $\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$ is equal to:

$\frac{2\sin A \quad 1+\sin A}{1+\sin A+\cos A}$ का मान किसके बराबर है?

a) 1+ sinA-cosA

b) 1-sinAcosA

c) 1+ cosA-sinA

d) 1+ sinAcosA

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(14) What is the value of $\sin(90° + 2A)[4-\cos^2(90° -2A)]$?

$\sin(90° + 2A)[4 - \cos^2(90° - 2A)]$ का मान क्या है?

a) $2(\cos^3A-\sin^3A)$

b) $2(\cos^3A+ \sin^3A)$

c) $4(\cos^6A+ \sin^6A)$

d) $4(\cos^6A-\sin^6A)$

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(15) (HW) If $x = cosec\theta - \sin\theta$ and $y = \sec\theta - \cos\theta$, then the relation between x and y is

यदि $x = cosec\theta - \sin\theta$ और $y = \sec\theta - \cos\theta$ हो, तो x और y के बीच सम्बन्ध होगा

a) $x^2 + y^2 + 3 = 1$

b) $x^2y^2 (x^2 + y^2 + 3) = 1$

c) $x^2 (x^2 + y^2 - 5) = 1$

d) $y^2 (x^2 + y^2 - 5) = 1$

(16) If $cot\,\theta + tan\,\theta = p$ and

(HW) $sec\,\theta - cos\,\theta = q$ then

$(p^2q)^{\frac{2}{3}} - pq^{2})^{\frac{2}{3}}$ =?

यदि $cot\,\theta + tan\,\theta = p$ और $sec\,\theta - cos\,\theta = q$ है, तो

$p^2q^{\frac{2}{3}} - pq^{2\frac{2}{3}}$ =?

a) 0 b) 1

c) 2 d) 3

CDS 2016 I

(17) If $\frac{\cos\alpha}{\sin\alpha+\cos\beta} + \frac{\cos\beta}{\sin\beta-\cos\alpha} = \frac{x}{\sin\alpha-\cos\beta} + \frac{\cos\beta}{\sin\beta+\cos\alpha}$ then 'x' is equal to:

यदि $\frac{\cos\alpha}{\sin\alpha+\cos\beta} + \frac{\cos\beta}{\sin\beta-\cos\alpha} = \frac{x}{\sin\alpha-\cos\beta} + \frac{\cos\beta}{\sin\beta+\cos\alpha}$ है, 'x' तो बराबर है:

a) $\cos\beta$ b) $\cos\alpha$

c) $\sin\beta$ d) $\sin\alpha$

SSC CHSL 2018

(18) What is the value of $\left[(sec\,2\theta + 1)\sqrt{sec^2\theta - 1}\right] \times \frac{1}{2}(cot\ \theta - tan\ \theta)$?

$\left[(sec\,2\theta + 1)\sqrt{sec^2\theta - 1}\right] \times \frac{1}{2}(cot\,\theta - tan\,\theta)$ का मान क्या है?

a) 0 b) 1

c) $cosec\ \theta$ d) $sec\ \theta$

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(19) If sinA+ sinB= C& cosA+ cosB= D, then sin(A+ B)=?

यदि $sinA + sinB = C$ और $cosA + cosB = D$ है, तो $sin(A + B) = ?$

a) CD

b) $\frac{CD}{C^2+D^2}$

c) $\frac{C^2+D^2}{2CD}$

d) $\frac{2CD}{C^2+D^2}$

20 If $\alpha + \beta = 90°$ then the value of $(1\text{-}\sin^2\alpha)(1\text{-}\cos^2\alpha) \times (1+\cot^2\beta)(1+\tan^2\beta)$ is

अगर $\alpha + \beta = 90°$ है, तो $(1\text{-}\sin^2\alpha)(1\text{-}\cos^2\alpha) \times (1+\cot^2\beta)(1+\tan^2\beta)$ का मान:

a) 1 b) -1

c) 0 d) 2

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HW

If $\angle A$ and $\angle B$ are complementary to each other, then the value of $\sec^2 A + \sec^2 B - \sec^2 A \sec^2 B$ is

अगर कोण $\angle A$ और $\angle B$ एक दुसरे के पूरक हैं, तो $sec^2 A + \sec^2 B - \sec^2 A \sec^2 B = ?$

a) 1 b) -1

c) 2 d) 0

22 HW The value of the expression $(\cos^6\theta + \sin^6\theta - 1)(\tan^2\theta + cot^2\theta + 2)$ is:

व्यंजक $(cos^6\theta + sin^6\theta - 1)(tan^2\theta + cot^2\theta + 2)$ निम्न में से किसके बराबर है?

a) 0 b) -1

c) -3 d) 1

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23 $\tan\left(\frac{\pi}{4}+\theta\right)-\tan\left(\frac{\pi}{4}-\theta\right)=?$

a) $2\tan 2\theta$ b) $2\cot 2\theta$

c) $\tan 2\theta$ d) $\cot 2\theta$

(24) If A, B, C are angles of a triangle, then $\sin^2 A + \sin^2 B + \sin^2 C - 2\cos A \cos B \cos C = ?$

यदि A, B, C एक त्रिभुज के कोण हैं, तो $sin^2 A + sin^2 B + sin^2 C - 2\cos A \cos B \cos C = ?$

a) 1 b) 2

c) 3 d) 4

(25) If A+ B+ C= $\frac{3\pi}{2}$, then cos2A+ cos2B+ cos2C=?

यदि $A + B + C = \frac{3\pi}{2}$ है, तो cos2A+ cos2B+ cos2C=?

a) 1-4sinAsinBsinC
b) 1-sinAsinBsinC
c) 1-2sinAsinBsinC
d) 1-4cosAcosBcosC